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INSID&S Lab
***Int**eraction and **S**emantics for*
***In**novation with **D**ata **&** **S**ervices*

ARTIFICIAL INTELLIGENCE @ UNIMIB

2.3. RDF in Action

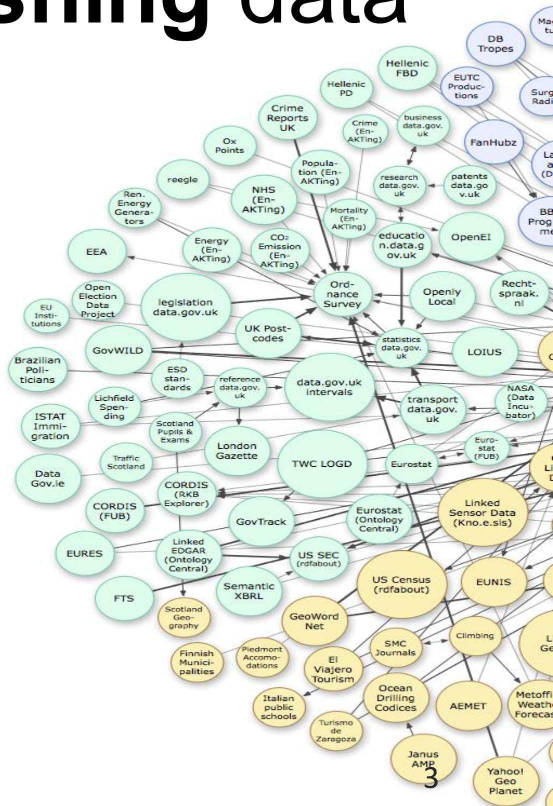
Linked Data and Semantic Annotations

Linked Data

A.k.a. best practices to publish and access Semantic Knowledge Graphs using open standards

Linked Data

- Set of best practices for **publishing** data on the Web.
- Data from different knowledge domains, self-described, linked and accessible.
- Follows 4 simple principles…



Linked Data Principles

1. Use IRIs as **names** for things.
2. Use HTTP IRIs so that users can **look up** those names.
3. When someone looks up a IRI, **provide <u>useful</u> information**, using the standards (RDF*, SPARQL).
4. Include links to other IRIs, so that users can **discover** more things.

Linked Data Principles

3. When someone looks up a IRI, **provide <u>useful</u> information**, using the standards (RDF*, SPARQL, Turtle).

What to return for a URI?

- **Immediate description:** triples where the URI is the subject.
- **Backlinks:** triples where the URI is the object.
- **Related descriptions:** information of interest in typical usage scenarios.
- **Metadata:** information as author and licensing information.
- **Syntax:** RDF descriptions as RDF/XML and human-readable formats.

Source: *How to Publish Linked Data on The Web* - Chris Bizer, Richard Cyganiak, Tom Heath.

Elton John Data in DBpedia

https://dbpedia.org/resource/Elton_John

DBpedia · Browse using · Formats

dbp:birthDate	• 1947-03-25 (xsd:date)
dbp:birthName	• Reginald Kenneth Dwight (en)
dbp:birthPlace	• Pinner, Middlesex, England (en)
dbp:caption	• John attending the premiere of The Union at the 2011 Tribeca Film Festival (en)
dbp:children	• 2 (xsd:integer)
dbp:footer	• Influenced by the glam rock scene in the UK, John often wore elaborate stage costumes, including platform boots . (en)
dbp:honorificPrefix	• dbr:Knight_Bachelor
dbp:image	• Elton John's shoes .jpg (en) • Elton john cher show 1975.JPG (en)
dbp:name	• Elton John (en)
dbp:occupation	• Singer (en) • (en) • composer (en) • pianist (en)

e.g., **links**: triples with the entity as subject

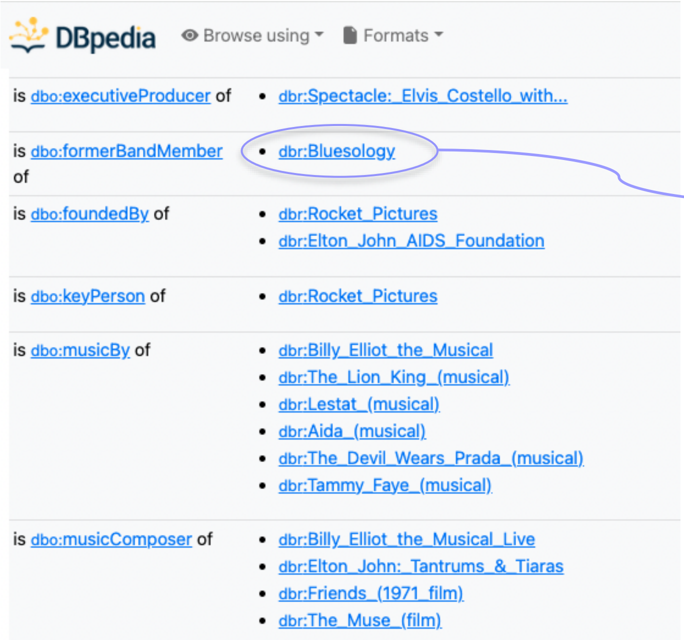
DBpedia · Browse using · Formats

is dbo:executiveProducer of	• dbr:Spectacle:_Elvis_Costello_with...
is dbo:formerBandMember of	• dbr:Bluesology
is dbo:foundedBy of	• dbr:Rocket_Pictures • dbr:Elton_John_AIDS_Foundation
is dbo:keyPerson of	• dbr:Rocket_Pictures
is dbo:musicBy of	• dbr:Billy_Elliot_the_Musical • dbr:The_Lion_King_(musical) • dbr:Lestat_(musical) • dbr:Aida_(musical) • dbr:The_Devil_Wears_Prada_(musical) • dbr:Tammy_Faye_(musical)
is dbo:musicComposer of	• dbr:Billy_Elliot_the_Musical_Live • dbr:Elton_John:_Tantrums_&_Tiaras • dbr:Friends_(1971_film) • dbr:The_Muse_(film)
is dbo:narrator of	• dbr:The_Road_to_El_Dorado
is dbo:openingTheme of	• dbr:Rocket_Man_(TV_series)
is dbo:presenter of	• dbr:IHeart_Living_Room_Concert_for_America

e.g., **backlinks**: triples with the entity as object

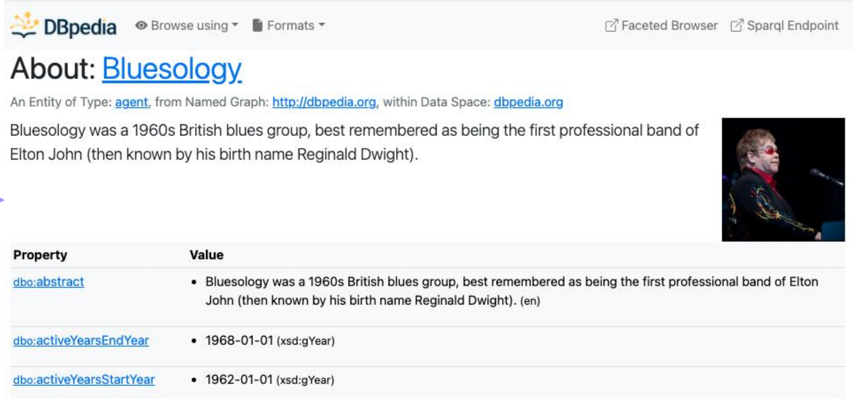
Linked Data Principles

4. Include links to other IRIs, so that users can **discover** more things.



DBpedia Browse using Formats

is dbo:executiveProducer of	• dbr:Spectacle:_Elvis_Costello_with...
is dbo:formerBandMember of	• dbr:Bluesology
is dbo:foundedBy of	• dbr:Rocket_Pictures • dbr:Elton_John_AIDS_Foundation
is dbo:keyPerson of	• dbr:Rocket_Pictures
is dbo:musicBy of	• dbr:Billy_Elliot_the_Musical • dbr:The_Lion_King_(musical) • dbr:Lestat_(musical) • dbr:Aida_(musical) • dbr:The_Devil_Wears_Prada_(musical) • dbr:Tammy_Faye_(musical)
is dbo:musicComposer of	• dbr:Billy_Elliot_the_Musical_Live • dbr:Elton_John:_Tantrums_&_Tiaras • dbr:Friends_(1971_film) • dbr:The_Muse_(film)



DBpedia Browse using Formats Faceted Browser Sparql Endpoint

About: Bluesology

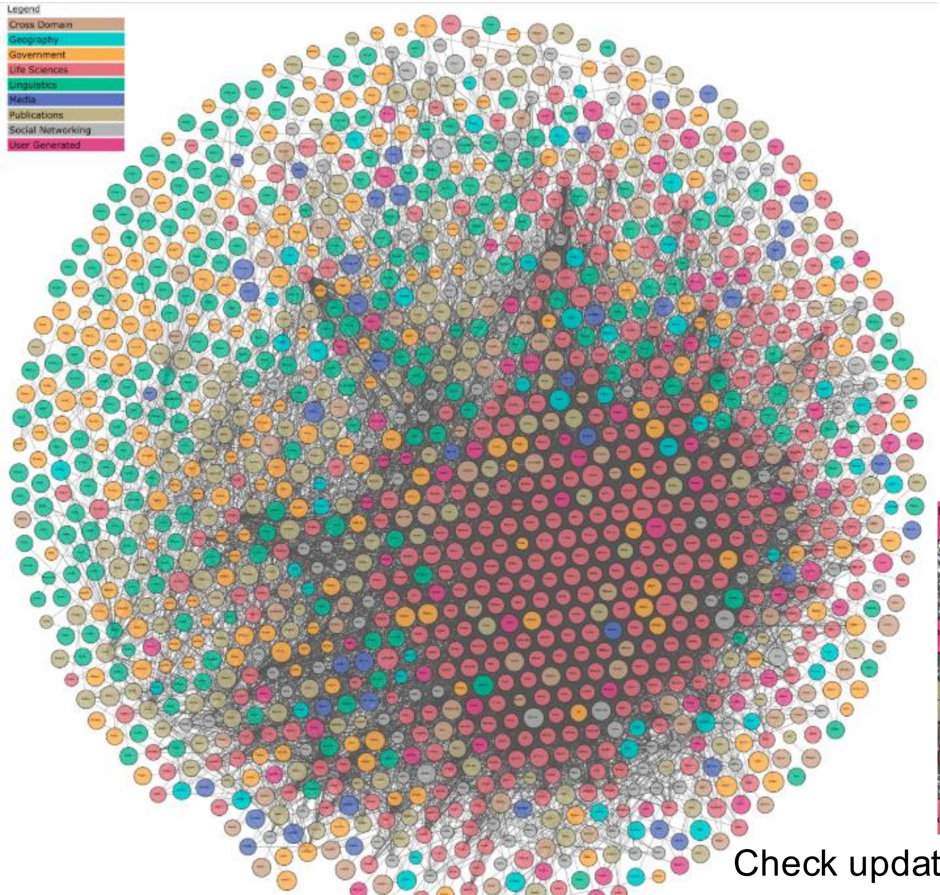
An Entity of Type: agent, from Named Graph: http://dbpedia.org, within Data Space: dbpedia.org

Bluesology was a 1960s British blues group, best remembered as being the first professional band of Elton John (then known by his birth name Reginald Dwight).

Property	Value
dbo:abstract	• Bluesology was a 1960s British blues group, best remembered as being the first professional band of Elton John (then known by his birth name Reginald Dwight). (en)
dbo:activeYearsEndYear	• 1968-01-01 (xsd:gYear)
dbo:activeYearsStartYear	• 1962-01-01 (xsd:gYear)

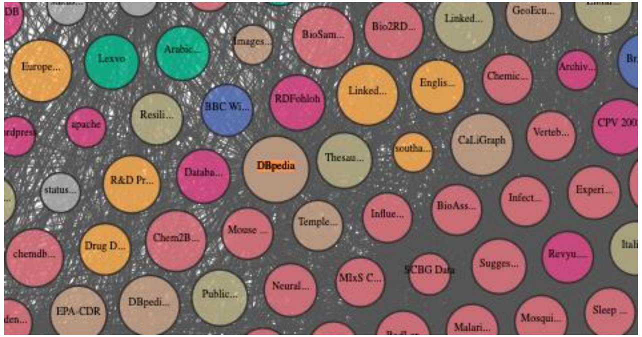
Links to other resources…
… the navigation continues

LOD Datasets on the Web



Latest stas
-1350 datasets
-9 topical categories

Previous stats
-80% of linking property is owl:sameAs

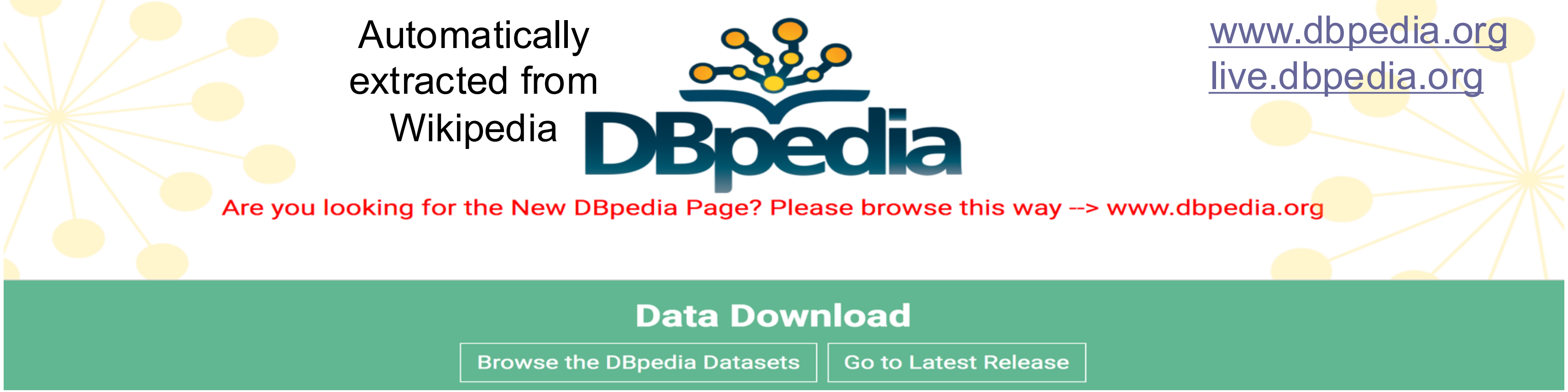


Check updates of figure and stats at: https://lod-cloud.net/

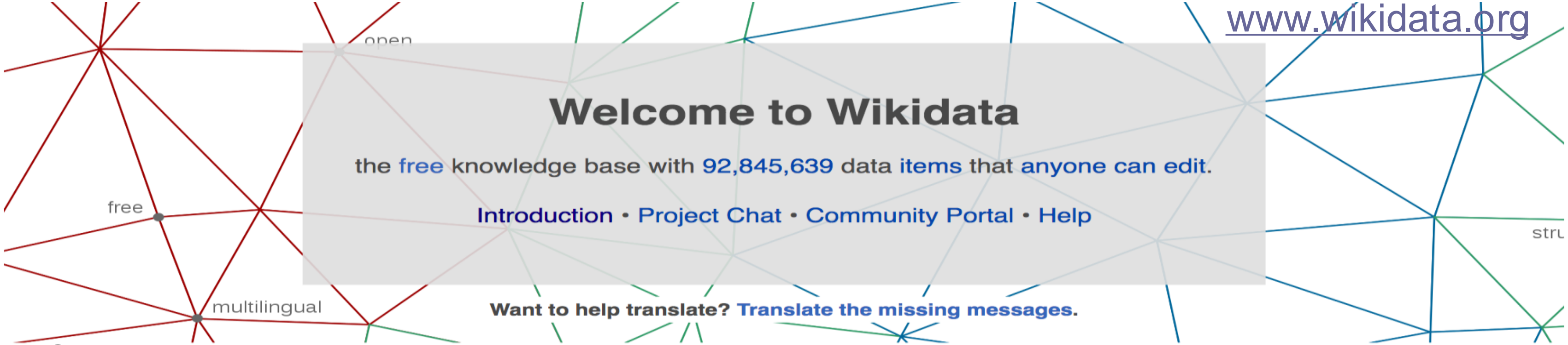
Two Important Hubs

Automatically extracted from Wikipedia

www.dbpedia.org
live.dbpedia.org



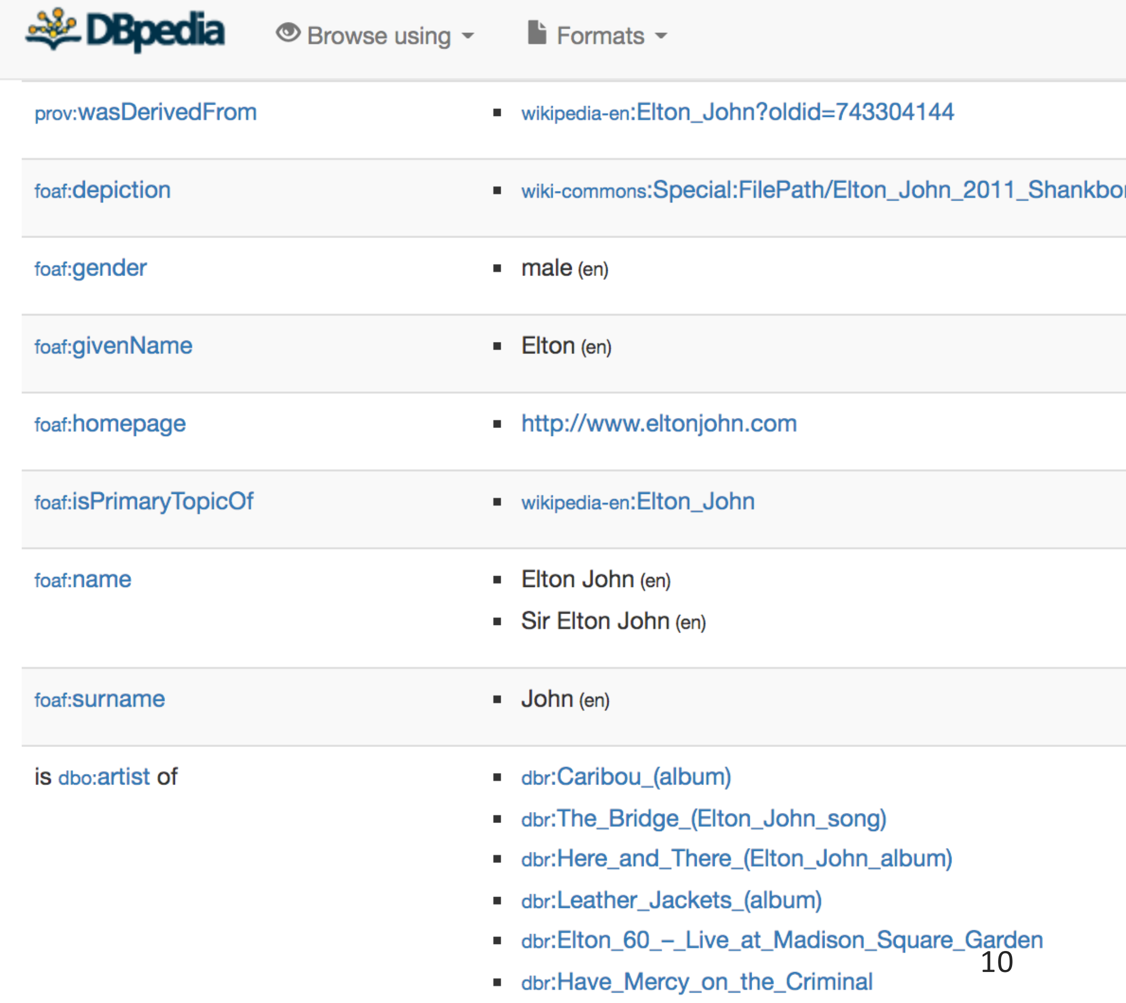
www.wikidata.org



Collaboratively built like Wikipedia

RDF Knowledge Graphs

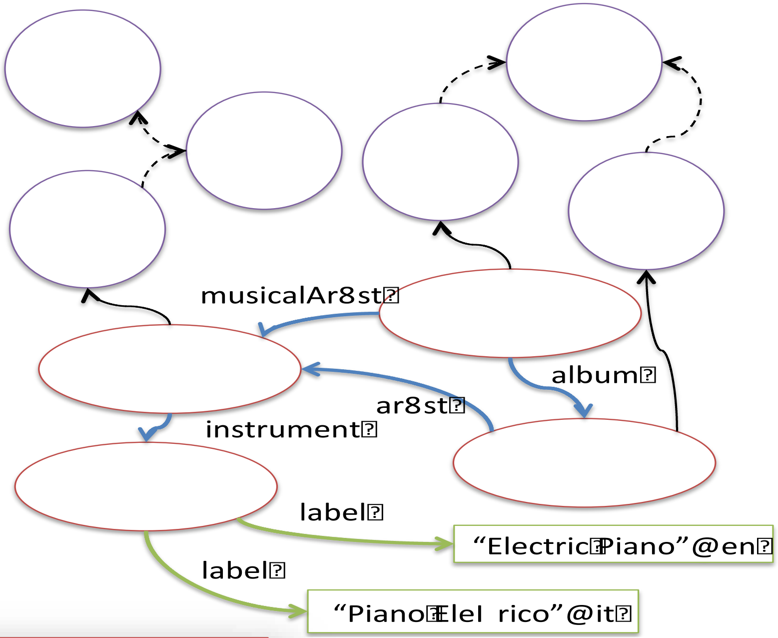
- Optimized for Knowledge Sharing on the Web
  - Publicly available KGs that can be accessed by third parties using Web protocols
  - Check the Linked Open Data Cloud
    - E.g., DBpedia, Wikidata


DBpedia   Browse using   Formats

prov:wasDerivedFrom	wikipedia-en:Elton_John?oldid=743304144
foaf:depiction	wiki-commons:Special:FilePath/Elton_John_2011_Shankbo
foaf:gender	male (en)
foaf:givenName	Elton (en)
foaf:homepage	http://www.eltonjohn.com
foaf:isPrimaryTopicOf	wikipedia-en:Elton_John
foaf:name	Elton John (en) Sir Elton John (en)
foaf:surname	John (en)
is dbo:artist of	dbr:Caribou_(album) dbr:The_Bridge_(Elton_John_song) dbr:Here_and_There_(Elton_John_album) dbr:Leather_Jackets_(album) dbr:Elton_60_–_Live_at_Madison_Square_Garden dbr:Have_Mercy_on_the_Criminal

"Big" Knowledge Graphs

- DBpedia (2015):
  - 6.2M entities (955K with geo coordinates)
  - 5M entities classified in an ontology (1.6M persons, 800K locations, 267K organizations)
  - 8.8B facts (triples)
    - E.g., < Barack Obama, birthplace, Honolulu>
- Ontology (schema):
  - 739 classes
  - ~1000 object properties
  - ~1600 datatype properties



<Electric Piano, label, "Electric Piano"@en>
<Electric Piano, label, "Piano Elel rico"@it>
<Elton John, instrument, Electric Piano>
<Empty Sky, ar8st, Elton John>
<Sails, musicalAr8st, Elton John>
<Sails, album, Empty Sky>
<Elton John, instrument, Electric Piano>
<Elton John, type, Musical Ar8st>
<Sails, type, Single>
<Empty Sky, type, Album>

<Musical Ar8st, subClassOf, Ar8st>
<Ar8st, subClassOf, Person>
<Album, subClassOf, Musical Work>
<Single, subClassOf, Musical Work>

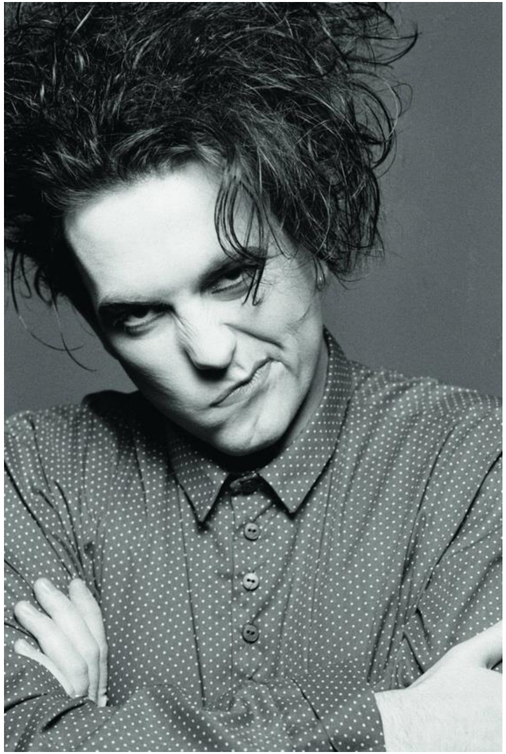
Find / Browse Quick Exercise

- DBpedia and Wikidata are indexed by Google (and other search engines)
  - E.g., try search for “elton john dbpedia” and “elton john wikidata” to get to the artist’s pages
- Quick exercise (write answers in the chat):
  - Find:
    - the band **Anderson Paak** is/was member of
    - all the record labels of Anderson Paak in DBpedia (live) and Wikidata and write which ones are not listed in both the sources
  - Browse to Aftermath Records and find:
    - its headquarters (city)

Remark: Don’t close the pages you find

"Create" Quick Exercise

- Create a graph to represent the essential content of this text passage:
  - "*The Cure is a band playing gothic rock. Popular songs of the band include Boys Don't Cry and Friday I'm in Love. Robert Smith is the leader and plays guitar*"
  - Try to use the examples in our presentations and the information collected during the previous exercise to find the right IRIs to use as predicates and types



Remark: Don't close the pages you find

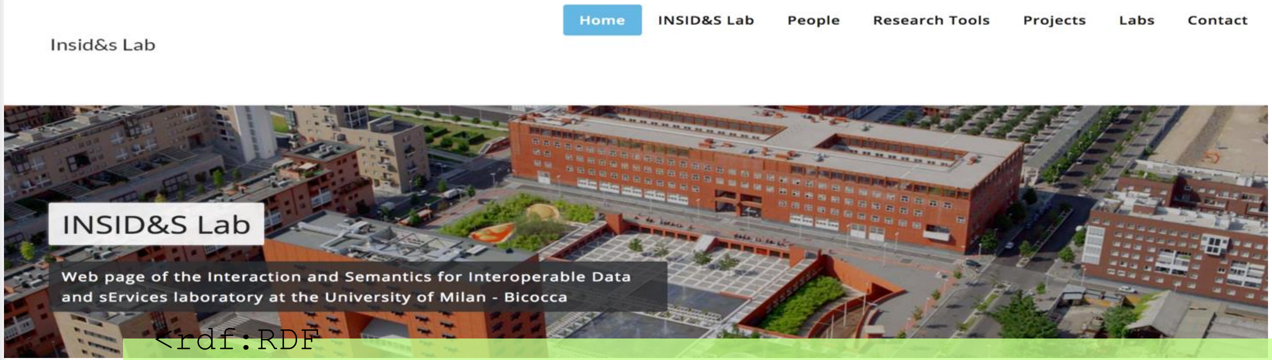
Semantic Annotations

Annotation of content in Web pages with semantics

The Original Idea of Semantic Web

- Use RDF as mark-up language to annotate content on the Web
- Example: describing *title* and *author* of a webpage

  (http://inside.disco.unimib.it/)



Idea:
Intelligent Agents can read this structured data, use it to query, filter and aggregate information

XML Syntax

```
<rdf:RDF
    xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
    xmlns:dc="http://purl.org/dc/elements/1.1/">
  <rdf:Description rdf:about="http://inside.disco.unimib.it/">
    <dc:title>Homepage of the INSID&S Lab at UNIMIB</dc:title>
    <dc:author>Marco Cremaschi</dc:author>
  </rdf:Description>
</rdf:RDF>
```

Annotation formats on the Web

Markup today

Results per Format

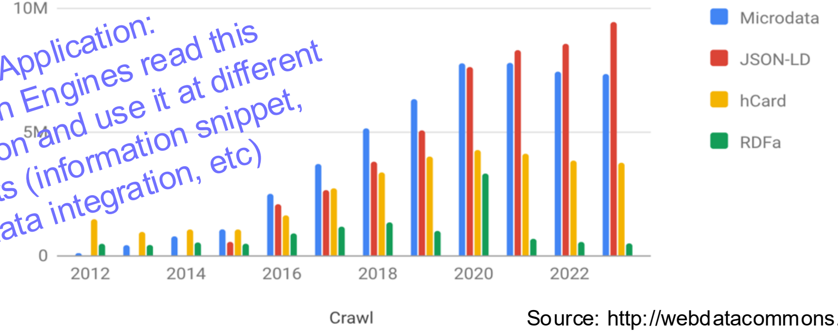
Format	Domains	URLs	Typed Entities	Triples
html-embedded-jsonld	8,596,990	877,812,654	9,388,554,696	45,840,052,993
html-microdata	7,471,628	801,909,298	6,030,281,759	28,824,001,037
html-mf-hcard	3,880,989	318,625,913	3,300,807,398	10,566,510,286
html-rdfa	594,018	91,100,238	275,563,980	768,304,041

Extraction Statistics

Crawl Date	October 2023
Total Data	98.38 Terabyte (compressed)
Parsed HTML URLs	3,353,090,410
URLs with Triples	1,696,953,312
Domains in Crawl	34,144,094
Domains with Triples	14,646,081
Typed Entities	20,890,660,670
Triples	97,689,391,384
Size of Extracted Data	1.8 Terabyte (compressed)

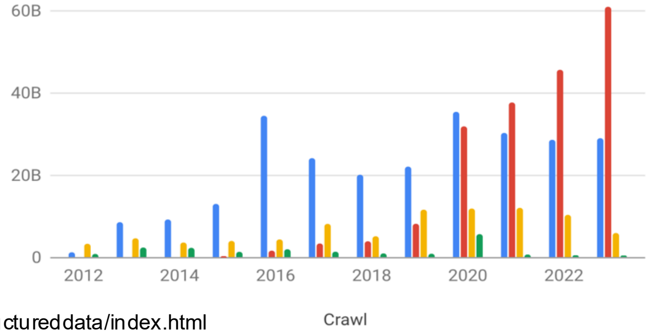
Number of PLDs Deploying the four Major Markup Formats

Pay-Level Domain (PLD) is a sub-domain of a public top-level domain, for which users usually pay for, where a single user or organization is likely to be in control



Application:
Search Engines read this information and use it at different extents (information snippet, data integration, etc)

Number of Triples marked up by the four Major Markup Formats

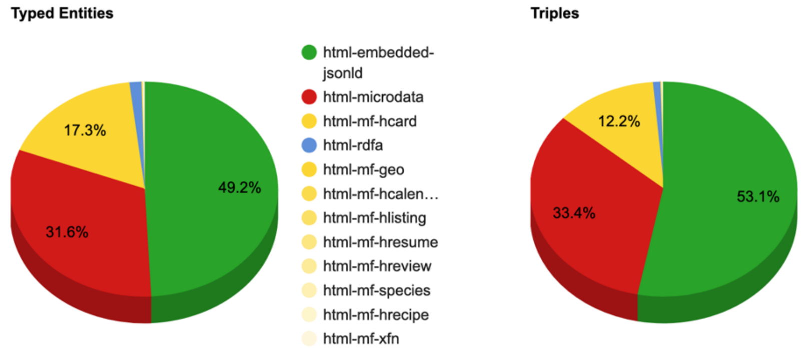


Source: http://webdatacommons.org/structureddata/index.html

Source: http://webdatacommons.org/structureddata/2022-12/stats/stats.html

Annotation formats on the Web

Markup today



Extraction Statistics

Crawl Date	October 2023	
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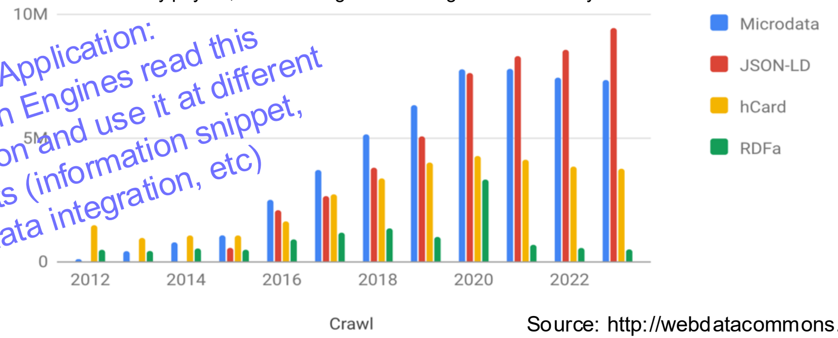

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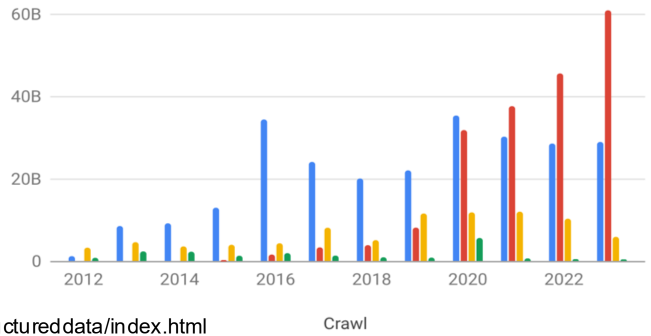

Source: http://webdatacommons.org/structureddata/2022-12/stats/stats.html

More Mark-up on the Web

- Mark-up today
  - Non RDF, not fully compatible but fairly transformable into RDF
    - Microdata, hCard
  - RDF or fully compatible
    - RDFa (native) + JSON-LD

More Mark-up on the Web

- Mark-up today
  - Non RDF, not fully compatible but fairly transformable into RDF
    - Microdata, **hCard**

Mix of hCard and vcard

```
<span class="vcard">
  <span class="fn">Max Mustermann<span>
  <div class="label" style="display:none">
    <span class="type">home</span>
    42 Plantation St.<br>Baytown, LA 30314<br>United States of America
  </div>
  <div class="adr">
    <span class="type">Home</span> Address:<br>
    <span class="street-address">42 Plantation St.</span><br>
    <span class="locality">Baytown</span>, <span class="region">LA</span>
    <span class="postal-code">30314</span><br>
    <span class="country-name">United States of America</span>
  </div>
</span>
```

Listing 2: Microformat Example 2

More Mark-up on the Web

- Mark-up today
  - Non RDF, not fully compatible but fairly transformable into RDF
    - **Microdata**, hCard

```
<section itemscope itemtype="http://schema.org/Person">
  Hello, my name is
  <span itemprop="name">John Doe</span>,
  I am a
  <span itemprop="jobTitle">graduate research assistant</span>
  at the
  <span itemprop="affiliation">University of Dreams</span>.
  My friends call me
  <span itemprop="additionalName">Johnny</span>.
  You can visit my homepage at
  <a href="http://www.JohnnyD.com" itemprop="url">www.JohnnyD.com</a>.
  <section itemprop="address" itemscope itemtype="http://schema.org/
      PostalAddress">
    I live at
    <span itemprop="streetAddress">1234 Peach Drive</span>,
    <span itemprop="addressLocality">Warner Robins</span>,
    <span itemprop="addressRegion">Georgia</span>.
  </section>
</section>
```

Listing 3: Microdata Example 1, Source: [8]

More Mark-up on the Web

- Mark-up today
  - Non RDF, not fully compatible but fairly transformable into RDF
    - Microdata, hCard
  - RDF or fully compatible
    - **RDFa** (native) + JSON-LD

```
<div vocab="http://xmlns.com/foaf/0.1/" typeof="Person"><p>
  <p>
    <span property="name">Alice Birpemswick</span>,
    Email: <a property="mbox" href="mailto:alice@example.com">alice@example.
        com</a>,
    Phone: <a property="phone" href="tel:+1-617-555-7332">+1 617.555.7332</a
        >
  </p>
</div>
```

More Mark-up on the Web

- Mark-up today
  - Non RDF, not fully compatible but fairly transformable into RDF
    - Microdata, hCard
  - RDF or fully compatible
    - RDFa (native) + **JSON-LD**

```
<script type="application/ld+json">
{
  "@context": "http://schema.org/",
  "@type": "Product",
  "name": "Lava Lite Classic Lava Lamp Purple/Blue",
  "aggregateRating": {
    "@type": "AggregateRating",
    "ratingValue": "3.5",
    "reviewCount": "60"
  },
  "offers": {
    "@type": "Offer",
    "priceCurrency": "USD",
    "price": "13.17",
    "availability": "http://schema.org/InStock"
  }
}
</script>
```

What You See / What is in the Webpage

Figure 1: Google search: Product display

```
<script type="application/ld+json">
{
  "@context": "http://schema.org/",
  "@type": "Product",
  "name": "Lava Lite Classic Lava Lamp Purple/Blue",
  "aggregateRating": {
    "@type": "AggregateRating",
    "ratingValue": "3.5",
    "reviewCount": "60"
  },
  "offers": {
    "@type": "Offer",
    "priceCurrency": "USD",
    "price": "13.17",
    "availability": "http://schema.org/InStock"
  }
}
</script>
```

JSON-LD

- JSON-LD is a JSON-based serialization for Linked Data
- JSON-LD offers a simple way to semantically augment JSON documents by adding *context* information and *hyperlinks* for describing the semantics of the different elements of a JSON objects.
- JSON-LD extends the JSON language with several keywords represented by the special names of JSON properties starting with the `@` sign.
- The three main reserved keywords the JSON-LD language adds to JSON are:

Key	Description	Example
`@context`	URL referencing a particular schema	http://schema.org/Person
`@id`	Unique identifier (usually a URI)	http://dbpedia.org/page/Mahatma_Gandhi
`@type`	A URL referencing the type of a value	http://www.w3.org/2001/XMLSchema#dateTime

https://json-ld.org/

JSON-LD: an example

JSON representing a Person

```
{
  "name": "Barack Obama",
  "givenName": "Barack",
  "familyName": "Obama",
  "jobTitle": "44th President of the United States”
}
```

JSON-LD identifying a person

```
{
  "@context": "http://schema.org",
  "name": "Barack Obama",
  "givenName": "Barack",
  "familyName": "Obama",
  "jobTitle": "44th President of the United States”
}
```

JSON-LD identifying a person (expanded)

```
[
  {
    "http://schema.org/name": [{"@value": "Barack Obama"}],
    "http://schema.org/givenName": [{"@value": "Barack"}],
    "http://schema.org/familyName": [{"@value": "Obama"}],
    "http://schema.org/jobTitle": [{"@value": "44th President of the United States"}]
  }
]
```

Best practice: JSON-LD representation
Barack Obama

Message of type Person retrieved from an API

```
{
  "@context": "http://schema.org",
  ”@id": " https://www.wikidata.org/wiki/Q76",
  "type": "Person",
  "name": "Barack Obama",
  "givenName": "Barack",
  "familyName": "Obama",
  "jobTitle": "44th President of the United States”
}
```

... with augmented values

Enumeration type {male, female}

```
{
  "@context": ["http://schema.org", {
    "gender": {"@id": "schema:gender", "@type": "@vocab"}
  }],
  "@context": "http://schema.org",
  "@id": "https://www.wikidata.org/entity/Q76",
  "type": "Person",
  "name": "Barack Obama",
  "givenName": "Barack",
  "familyName": "Obama",
  "jobTitle": "44th President of the United States”,
  "gender": "male"
}
```

JSON-LD for Things

- We can use the Product schema described on `schema.org` to add some semantic data to device descriptions (e.g., `pi.json` for Pi device)

`resources/piJsonLd.json`:

```
{
  "@context": "http://schema.org/Product",
  "@id": "http://localhost:8484",
  "name": "My WoT Raspberry PI",
  "description": "A simple WoT-connected Raspberry PI for the WoT book.",
  "productID" : "asin:B00T2U7R7I",
  "manufacturer" : "Raspberry Pi",
  "model" : "100437",
  "image" : "https://www.raspberrypi.org/wp-
  content/uploads/2015/01/Pi2ModB1GB_-comp-500x283.jpeg",
  [...]
```

A link to the vocabulary for describing a product, available on schema.org

The URL identifier for this web Thing

References an image for the Pi

name, description, productID, manufacturer, and model have an agreed-upon meaning in a schema.org product context.

- All clients that understand the `http://schema.org/Product` context will be able to automatically process this information in a meaningful way

Source: https://www.manning.com/books/building-the-web-of-things

Search / Find Quick Exercise

- Search for a semantic annotation of a product hidden in some web page
  - Hints:
    - write a search that points to a product (e.g., “omega watches”)
    - find some result that looks like “more structured”
    - follow the link to the webpage
    - inspect the webpage code and search for annotations

Search / Find Quick Exercise / Sol.

omega watches

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Annuncio · www.omegawatches.com/

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https://www.gioielleriacasella.com/14752-orologio-omega-de-ville-prestige-co-axial-con-quadrante-argento-e-cinturino-in-pelle-nero-7612586231124.html

Annunci · Acquista ora

New Fashion TUDOR Pelag...
80,00 €
Wish
Da Google

Omega orologio Omega de vill...
3.400,00 €
Gioielleria Casel
Spediz. gratuita
Da Drezzy.it

OMEGA Speedmaster...
5.900,00 €
Omega Watches
Spediz. gratuita
Da Google

GIOIELLERIA Casella

Cerca un prodotto

TUTTI I PRODOTTI · DONNA · UOMO · IDEE REGALO · BRANDS · CONSULENZA E ASSISTENZA

Home / Orologi / Orologi automatici / Orologio Omega De Ville Prestige Co-Axial con Quadrante Argento e Cinturino in Pelle Nero

Orologio Omega De Ville Prestige Co-Axial con Quadrante Argento e Cinturino in Pelle Nero

RIFERIMENTO: 42413402002001

3.400,00 €

Tasse incluse


```
<!-- SCHEMA.ORG prodotti SCHEDE PRODOTTO -->
        <script type="application/ld+json">{
    "@context":"https://schema.org/",
    "@type":"Product",
    "name":"Orologio Omega De Ville Prestige Co-Axial con Quadrante Argento e C
    "description":" Gli orologi  De Ville Prestige di Omega  sono divenuti una
Questo modello è caratterizzato da una elegante  cassa in acciaio da 39,5   mm  r
Lo straordinario quadrante, con  datario  a ore 3 e lancette argentate, è protett
Corredato di un  cinturino in pelle nero  è animato da un  calibro   automatico C
Un&#039;orologio elegante di grande impatto visivo. ",
    "SKU":"42413402002001",
    "MPN":"42413402002001",
    "image":[
      "https://www.gioielleriacasella.com/12161-large_default/orologio-omega-d
    ],
    "brand":{
      "@type":"Thing",
      "name":"Omega"
    },
    "offers":{
      "@type":"Offer",
      "url":"https://www.gioielleriacasella.com/14752-orologio-omega-de-ville-
      "priceCurrency":"EUR",
      "price":"3400.00",
      "itemCondition":"https://schema.org/NewCondition",
      "availability":"https://schema.org/InStock",
      "category":"Orologi automatici",
      "seller":{
        "@type":"Organization",
        "name":"Gioielleria Casella S.r.l.",
        "logo":"https://www.gioielleriacasella.com/gioielleria-casella.jpg"
      }
    }
  }</script>
```

More Resources

- RDF:
  - https://www.w3.org/RDF/ (ultra-short intro in the w3 site with relevant links)
  - Intro to RDF*: https://www.w3.org/TR/rdf11-primer/
  - Syntax (turtle)*: https://www.w3.org/TR/turtle/
- JSON-LD:
  - Intro & syntax:

https://www.w3.org/TR/2020/CR-json-ld11-20200305/

*w3C documents are always good resources to refer to when working with a language. They are long but well-structured so it's not too difficult to find the information you search for

Summary

- RDF: triple-based data model (IRIfied Graphs + BN + Literals)
- Linked Data: make data available and interconnected
- Semantic annotations in web pages: “structured” content for search engines